

AI & Machine Learning - Task 2

Feature Engineering, Model Optimization & Performance Comparison

Objective

Compare multiple machine learning models using the California Housing Dataset and identify the best-performing model.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-Learn
- Matplotlib
- Joblib

Models Implemented

1. Linear Regression
2. Ridge Regression
3. Decision Tree Regressor

Evaluation Metrics

- RMSE (Root Mean Squared Error)
- R² Score

Results

Model	RMSE	R ² Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Decision Tree Regressor	0.724234	0.599732

Best Model

Decision Tree Regressor

Files Included

- AI_ML_Task2_Model_Comparison.ipynb
- Task2_Report.pdf
- best_model.pkl

Internship

Completed as part of the AI & Machine Learning Internship at Maincrafts Technology.